
Executive Report

RealEstate Hub AI Voice Agent

1. Executive Summary

This agent is a production-oriented, Urdu/English (“UrduLish”) AI voice agent built for RealEstate Hub. It handles inbound property inquiries, gives recommendations grounded in a real listings database, and manages the full appointment lifecycle (booking, rescheduling, cancellation) with live Google Calendar and Gmail integration. Every conversation is logged to a CRM data store for follow-up and reporting.

The system has moved from working prototype to a hardened, evaluated build: a 44-scenario production test suite, a dedicated security/prompt-injection suite, latency and reliability monitoring, containerized deployment, and CI checks are all in place. A telephony front end (VAPI with a live phone number) is operational for real inbound calls, alongside text and microphone test interfaces that share one conversation engine.

This report summarizes the project's objectives, technical architecture, evaluation results, known limitations, and the roadmap for the next phase. Its purpose is to give stakeholders a clear, honest picture of what is production-ready today, what is still in progress, and what investment is needed to scale further.

2. Objectives

The project set out to answer a practical question: can an AI voice agent handle real, unscripted customer calls for a real estate business, in the customer's natural mixed Urdu/English speech, accurately enough and reliably enough to be trusted with production traffic?

2.1 Primary goals

- Automate first-line customer engagement: answer property questions, understand buyer/seller/renter/investor intent, and recommend relevant listings without human intervention.
- Automate the appointment lifecycle end to end: booking, rescheduling, and cancellation, synced to a real calendar so double-booking cannot occur.
- Communicate naturally in UrduLish, the way target customers actually speak, rather than forcing a single-language interaction.
- Ground every factual claim (prices, availability, appointment confirmations) in real data and real API results, never in model guesswork, to prevent hallucinated commitments.
- Log every interaction to a CRM so sales staff retain full visibility and follow-up capability.
- Build the underlying pipeline to be domain-agnostic, so the same architecture can be redeployed for another business vertical by changing configuration, data, and prompts only.

2.2 Success criteria

- Correct intent routing and recommendation continuity across multi-turn conversations.
- Reliable, race-condition-free calendar booking (“fail closed” rather than double-book).
- Resilience against prompt injection, jailbreak attempts, and requests for unauthorized internal data.
- Acceptable voice latency for a natural phone conversation.
- A deployable, monitorable, containerized production build.

3. System Architecture

Voice agent is built around a single stateful conversation engine (a LangGraph orchestration graph) that all front ends share. This “one brain, many mouths” design keeps conversation logic in one place and lets input/output channels vary independently.

3.1 Core components

- **API layer (FastAPI):** entry point for client applications; exposes health/readiness probes, monitoring metrics, and the conversation-turn endpoint.
- **Orchestration layer (LangGraph):** manages stateful conversation flow and routes intent — small talk, recommendation, booking, rescheduling, cancellation — while preserving session context (caller identity, property preferences, in-progress appointment drafts).
- **Security guard:** a deterministic, pre-LLM filter that intercepts prompt injection, jailbreak attempts, and unauthorized data requests before they ever reach the language model.
- **Retrieval-Augmented Generation (RAG) engine:** ChromaDB vector store over the real property catalogue, using a local sentence-transformer embedding model, so recommendations and factual answers are grounded in real listings rather than invented by the LLM.
- **Tool integrations:** a SQLite-backed CRM for customer profiles and interaction history; Google Calendar for real-time availability and event creation (built to “fail closed” to prevent double-booking); Gmail for internal notification emails on bookings/cancellations; Deepgram for speech-to-text and Fish Audio/ElevenLabs for text-to-speech.

3.2 Front ends over one shared brain

- **Text conversation interface;** used for fast functional testing without any audio code.
- **Live microphone/speaker pipeline :** Deepgram live STT → conversation engine → Fish Audio TTS, for local voice testing.
- **Telephony integration (VAPI, with a live phone number provisioned):** Deepgram Nova 3 STT, the conversation engine as a custom LLM backend, and ElevenLabs TTS, so real phone calls are routed to the same underlying logic used in testing.

Because conversation logic lives only in the orchestration layer, none of the front ends contain business logic of their own, they are I/O adapters. This was an explicit design constraint to keep the system maintainable as more channels (e.g. WhatsApp) are added.

3.3 Deployment

- Containerized via Docker, with persistent volumes for the SQLite database, vector store, and credentials.
- docker-compose for local/staging orchestration; a GitHub Actions CI pipeline runs syntax checks, deterministic regression tests, and a production image build on every change.
- Credentials (Google service account, API keys) are mounted at runtime, never baked into the image.

4. Evaluation

Rather than relying on informal spot-checks, the team built a structured, repeatable production evaluation suite covering realistic call scenarios, adversarial security attacks, and performance/latency measurement.

4.1 Conversation evaluation suite

A 44-scenario suite exercises 11 realistic caller categories:

Category	Scenarios	Category	Scenarios
Buyer	5	Rescheduling	3
Seller	4	Off-topic	3
Investor	4	Prompt injection	6
Rental	4	Angry customer	4
Appointment	5	Silent caller	3
Cancellation	3	Total	44

Checks cover intent routing accuracy, recommendation continuity, booking-state continuity, appointment completion, rescheduling and cancellation correctness, multi-turn memory retention, off-topic handling, prompt-injection resistance, and behavior under angry or silent callers.

The most recent recorded run of this suite scored 50 of 52 checks passing across the 44 scenarios (0 scenario-level errors). The two failing checks were real, specific bugs:

A negation-handling gap where a buyer explicitly rejecting a property type (e.g. “apartment chahiye, godam nahi”) was still scored as a positive match for the rejected type.

- A keyword-only appointment detector that missed an implicit reschedule request (a caller proposing a new date/time for an existing appointment without saying the literal word “reschedule”).

Both fixes are implemented and covered by a new, LLM-independent regression test suite, but the full 44-scenario suite has not yet been re-run end-to-end against the fixed code in a fully credentialed environment, this re-run is the immediate next verification step, noted honestly here rather than assumed.

4.2 Security / prompt-injection testing

A dedicated adversarial suite tests four attack families: instruction-override (“ignore your instructions”), system-prompt extraction, fabricated booking confirmations, and requests for internal/confidential company data: 16 variants in total, plus 6 core attacks folded into the main scenario suite. The deterministic security guard intercepts these before they reach the LLM, and this layer runs independently of model behavior, which reduces reliance on the LLM “behaving well” under attack.

4.3 Performance evaluation

A performance harness measures, per turn: mean/p50/p95/max latency, conversation and booking success rate, tool failure rates (calendar, email), RAG retrieval accuracy against 10 factual FAQ cases, multi-turn memory accuracy, and a hallucination/grounding proxy that flags unsupported numeric claims or answers that miss the expected grounded fact. This proxy is explicitly labeled as a proxy: final semantic hallucination scoring still requires human review, and that review has not yet been completed.

4.4 Monitoring in production

A metrics layer records turn latency, API latency, STT confidence, TTS first-byte latency and failures, calendar/email failures, booking success, and RAG misses, exposed via a metrics endpoint. Target SLAs have been defined for production operation:

Metric	Target
STT latency	< 300 ms
Orchestration latency (standard turn)	< 1500 ms
Orchestration latency (RAG/Calendar turn)	< 2500 ms
TTS first-byte latency	< 400 ms
End-to-end P95 turnaround	< 2200 ms
Core API / orchestration uptime	99.9%

5. Limitations

The following gaps and risks are flagged explicitly so stakeholders can weigh them against the roadmap, rather than assuming full production maturity.

- The full 44-scenario and prompt-injection suites need a fresh end-to-end re-run against the current, fixed codebase in a properly credentialed environment (LLM, Deepgram, Google Calendar/Gmail, TTS). The most recent saved results predate some of the source fixes described in this report and should not be treated as current evidence.

- The CRM is a local SQLite database, adequate for the pilot but not yet integrated with an enterprise CRM (e.g. Salesforce, HubSpot) that sales teams would use day to day.
- Property data is refreshed manually into the vector store rather than synced live from an MLS or external listings feed; stale inventory is possible if refresh discipline lapses.
- Language coverage is Urdu/English only; Punjabi and other regional languages are not yet supported.
- WhatsApp and SMS channels are not yet built: today's channels are the phone call and text/voice test interfaces only.
- Backup, credential-rotation, and log-review procedures are documented and scheduled but depend on operational follow-through (cron jobs, weekly reviews) that has not yet accumulated a production track record.

6. Future Roadmap

Proposed phases are sequenced by dependency and expected impact. Near-term items close verification gaps; medium- and longer-term items expand reach and enterprise integration.

6.1 Near term: verification & hardening

- Re-run the full 44-scenario suite and the prompt-injection suite end-to-end against the current codebase in a credentialed environment; publish refreshed pass/fail numbers.
- Complete a human review pass to convert the hallucination proxy into a validated accuracy figure.
- Bring the VAPI telephony path to full logging and monitoring parity with the tested voice pipeline; load-test under realistic concurrent call volume.

6.2 Phase 2: omnichannel communication

- WhatsApp Business API integration, allowing property images, floor plans, and map locations to be shared directly in-chat.
- Automated SMS confirmations for bookings, reschedules, and cancellations to reduce no-shows.

6.3 Phase 2/3: enterprise ecosystem integration

- Migrate CRM writes from local SQLite to a direct Salesforce or HubSpot integration for centralized lead and call-summary access.
- Connect the vector store to a live MLS or external property feed to remove manual re-indexing and guarantee up-to-date inventory.

6.4 Phase 3: advanced capability & growth

- Extend language support to Punjabi and other regional languages to broaden addressable market reach.
- Evaluate branded voice cloning (e.g. custom TTS voices) for a more personalized brand experience.

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- Automatic follow-up campaigns for inquiries that did not convert to a booking.
 - Lead scoring (Hot/Warm/Cold) from transcript signals, with high-value leads routed to human closers.
 - Secure in-call payment links for booking fees or token money.
 - A stakeholder-facing analytics dashboard (building on the existing CRM dashboard) surfacing call volume, intent-routing accuracy, peak call times, and conversion rates in real time.

7. Recommendation

The core engine, data grounding, security layer, and appointment automation are functionally complete and have been evaluated against a structured, realistic test suite with a strong pass rate and transparently documented fixes. The recommended path to a confident production launch is: (1) complete the pending end-to-end suite re-run and hallucination human-review pass described in Section 5, and (2) validate the telephony channel under real call volume, before scaling marketing or call volume commitments. The Phase 2/3 roadmap can proceed in parallel without blocking this core launch.